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Syntax parsing model of Kazakh simple sentences

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ABSTRACT
This paper proposes a syntactic analysis of Kazakh simple sentences taking into account their semantics. To do this, first, the syntactic rules of sentences are described using formal grammar, then parsing trees and ontological models are built to determine the semantics of their components and the relationships between them. As a formal grammar used Chomsky's context-free grammar, and ontological models were built in the environment of Protége.

CCS Concepts
• **Computing methodologies→Natural language processing**

Keywords
Simple sentences; Formal grammar; Parsing of Kazakh language; Syntactic rules of sentences; Parsing tree; Ontological model.

1. INTRODUCTION
Syntax parsing is an important task in natural language processing. The main goal of syntactic analysis is the analysis of sentence structure.

Syntax analysis (parsing) is a main area of research in computational linguistics. Syntax is a section of linguistics that studies the ability, methods and schemes of the relationship of words in a sentence and the principles of constructing a sentence, their structure. The sentences structure and their types are closely related to morphology. Words are part of a phrase or sentence, and endings are a morphological-syntactic category of combining words with each other [15 - 22].

Parsing (syntax analysis or syntactic analysis) is a process of comparing the linear sequence of tokens (words, tokens) of a natural or formal language with its formal grammar.

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The result of parsing is the syntactic structure of the sentence, represented either as a dependency tree or as a constituency tree.

Syntax defines the rules exploited to organize the words in a sentence on the basis of the following elements: constituents, grammatical relations, subcategorizations and dependency relations [1].

Constituents are groups of words that may behave as a single unit or phrase. Here, the constituents are described using parts of speech and phrase. The noun phrase consists of groups made up of nouns, pronouns, adjectives, numerals (my house, large room).

Grammatical relations are a formalization of sentence structure as a relationship between subjects and objects and other notions. The constituents are characterized by their role in the sentence and the parts of sentences are determined. In the Kazakh language there are five parts of the sentences: Subject, Predicate, Attribute, Adverbial, Object. For example, ol [SUBJECT] kitapty [OBJECT] aldy [Predicate](He took the book).

Subcategorizations and dependency relations refer to certain kinds of relations between words and phrasal groups. For example, want can be followed either by a verb infinite form or a noun phrase as object (I want to walk, I want a tea) whereas find can be followed only by a noun phrase (I found a key).

There are a variety of techniques used for phrase recognition using rule-based technique, statistical technique, and their combination [2], [3], [4], [16], [17], [19]. The paper [5] describes solution for identifying Kazakh basic phrases using rule-based and the maximum entropy methods. The recognition of basic Kazakh phrase could simplify sentence structure, reduce the difficulty of syntactic analyzer. The [6] describes the work on parsing Turkish using the lexical-functional grammar formalism. The implementation is based on Tomita's parser. The researchers proposed a number of analyses methods for natural language sentences [7], [8], [9], [21]. In this work, we will describe the syntax of the Kazakh language using the Constituency grammar. We use context-free grammar to formalize the syntax of Kazakh that allow modeling these constituencies.

2. MODELING KAZAKH SENTENCES STRUCTURE

For modeling syntax of natural languages are used constituency grammars and dependency grammar in terms of a hierarchically ordered structure of constituents their sentences [1]. Each and every word usually belongs to a specific lexical category in the case and forms the head word of different phrases [10].

Context free Chomsky grammar (CFG) is the most widely used formal system for modeling constituent structure in natural languages. CFG consists of a set of rules or productions, each of which expresses the ways that symbols of the language can be grouped and ordered together, and a lexicon of words and symbols. CFG G is defined by four parameters [11]:

$$G = < N_s, S, T_s, R >,$$

where N_s – a set of nonterminal symbols; $S \in N$ – a start nonterminal symbol; T_s – a set of terminal symbols; R – a set of rules of the form $A \rightarrow \alpha$, $A \in N$ – a nonterminal symbol, $\alpha \in (N_s \cup T_s,)^*$ – a string of of symbols from the infinite set of strings $(N_s \cup T_s,)*$.

There are 20 types of simple narrative sentences in the Kazakh language. In [12, 13], these structures were described using the following CFG rules:

S → Sub | Sub Pre | Sub Obj Pre | Sub Adl Pre |Sub Obj Adl Pre|
Sub Adl Obj Pre | Sub Atr Obj Pre |Obj Atr Sub Pre | Obj Adl Sub Pre |
Obj Sub Adl Pre | Adl Sub Obj Pre |Atr Sub Obj Pre|
| Atr Sub Adl Pre | Sub Adl Atr Obj Pre |
Sub Atr Obj Adl Pre | Adl Sub Atr Obj Pre | Adl Obj Atr Sub Pre |

Atr Sub Adl Obj Pre | Atr Sub Obj Adl Pre | Atr Obj Sub Adl Pre,

Sub → N | Pron | Adj | Nom | Part | V_Inft | Intrj | Adv,

Pre → V | Adv | N | Nom | Pron | Imit_Auxv,

Obj → Abl_N | Abl_Pron | Abl_Adj | Abl_Num| Abl_Part |
Acc_N | Acc_Pron | Acc_Adj | Acc_Num| Acc_Part |
Dir_N | Dir_Pron | Dir_Adj | Dir_Num| Dir_Part |
Inst_N | Inst_Pron | Inst_Adj | Inst_Num| Inst_Part |
Loc_N | Loc_Pron | Loc_Adj | Loc_Num| Loc_Part,

Atr → Adj| N | Num | Pron | Part | Gen_Adj | Gen_N |
Gen_Num | Gen_Pron | Gen_Part,

Adl → Adv| N | Num | Pron | AdvPart | Imit | Adj |
Abl_N | Abl_Pron | Abl_Adj | Abl_Num| Abl_Part |
Dir_N | Dir_Pron | Dir_Adj | Dir_Num| Dir_Part |
Inst_N | Inst_Pron | Inst_Adj | Inst_Num| Inst_Part |
Loc_N | Loc_Pron | Loc_Adj | Loc_Num| Loc_Part

The various abbreviations used for these rules are presented in Table 1.

Table 1. List of abbreviations

Abbreviations	Abbreviations Meaning
Sub	Subject
Pre	Predicate
Obj	Object
Adl	Adverbial
Atr	Atribute
Adj	Adjective
Adv	Adverb
Imit	Imitative words
Num	Numeral
N	Noun
Pron	Pronoun
V	Verb
Part	Participle
Inter	Interjection
Conj	Conjunction
NP	Noun Phrase
S	Sentence
VP	Verb Phrase
Imit	Imitative words
Auxv	Auxiliary verb
Gen	Genitive case
Acc	Accusative(initial) case ending
Dir	Direction-dative case (directive) ending
Abl	Ablative case ending
Inst	Instrumental case ending
Loc	Localtive case ending

The structure of a simple sentence can be described using a grammar of phrases as follows:

S → NP VP

NP
→ N | N N | Pron | Num | N Pron | N Num | N Adv | N Adj | N Conj |
Pron N | Num N | Adv N | Adj N | Part N |
Pron Num | Num Pron | Pron Adv | Adv Pron | Pron Adj | Adj Pron |
Num Num | Num Adj | Adj Num | Adv Num |
Part Pron | Part Num | Part Adj |
Pron N Num Adv | Adj Adj Pron Pron |

VP
→ VP | V | V V | NP V | N V | Adv V | Adj V | Pron V | Imit V | Part V | Num V | C

For a formal description of the structure of simple sentences of the Kazakh language using the grammar of the components, specific parameter values must be specified. For example, let specific simple sentences of the Kazakh language be given:

For a formal description of the structure of simple sentences of the Kazakh language using the grammar of the components, specific parameter values must be specified. For example, let specific simple sentences of the Kazakh language be given: “Apasynan keshke habar keldi” - “News came from grandmother in the evening”.

Then, in order to formalize the syntactic rules of these simple

sentences, the CFG parameters will have the following values:

$N_s = \{S, N, V, NP, VP,\ Adv\}$;

$T_s = \{A, a, b, d, e, , h, i, k, l, n,\ p, r, s, y\}$;

$R = \{S \rightarrow NP\ VP, NP \rightarrow N \mid Adv,\ VP \rightarrow NP\ VP \mid N\ V\}$.

Parse tree for this simple sentence of the Kazakh language shown in Figure 1.

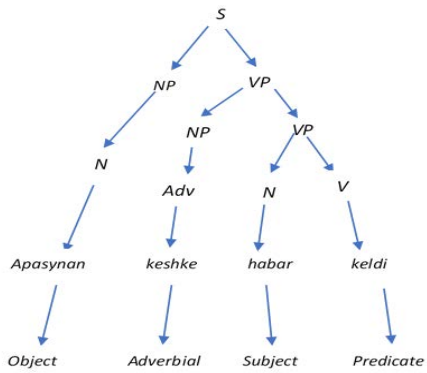

Figure 1. A parse tree "Apasynan keshke habar keldi".

3. ONTOLOGICAL MODELING OF SYNTACTIC RULES OF KAZAKH SIMPLE SENTENCE

Using the above grammar rules, the constructed ontological models of a simple narrative sentence of the Kazakh language will be shown in Figures 2-5.

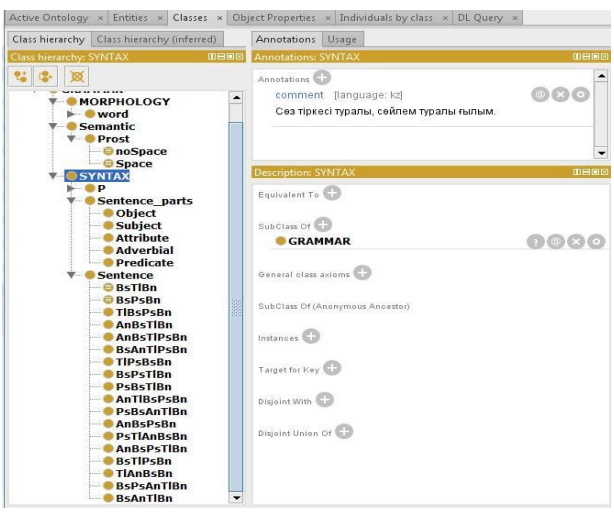

Figure 2. A fragment of the structure of the ontological model of a simple sentence of the Kazakh language.

Figure 4 shows the type of proposal BsTlBn (Bastauysh - Subject, Tolytauysh - Object, Bayandauysh - Predicate). For this type of sentence, the following necessary and sufficient conditions must be met:

$$BsTlBn \equiv \exists hasNP(PL\ or\ Pers) \sqcap \exists hasVP\ (DirVP1 \sqcap \left(\exists hasHead\ \left(V \sqcap \left(\exists hasRoot(root\ \sqcap (\forall isSpace\ noSpace))\right)\right)\right))$$

where: BsTlBn - Types of sentences consisting of Bastauysh - Subject, Tolytauysh - Object, Bayandauysh - Predicate;

hasNP - has a phrase;
PL - words in the plural;
Pers - personal pronouns;
hasVP - has a verb phrase;
DirVP1 - verb management (noun in the directive case + verb)
hasHead - has the main word;
V - verb;
hasRoot –has a root;
root - root
isSpace –is a word with spatial meaning;
noSpace – non-spatial semantics.

If all conditions are met, then when starting the resonator in Protégé, the sentence "Balalar tamaqqa toydy" - "Children ate food", which is an individual of the Sentence concep, is defined as the type of sentence BsTlBn, since the verb "toy" - "ate" is a non-spatial verb (Figure 3), and there will also be the necessary and sufficient conditions of this type of sentence are satisfied.

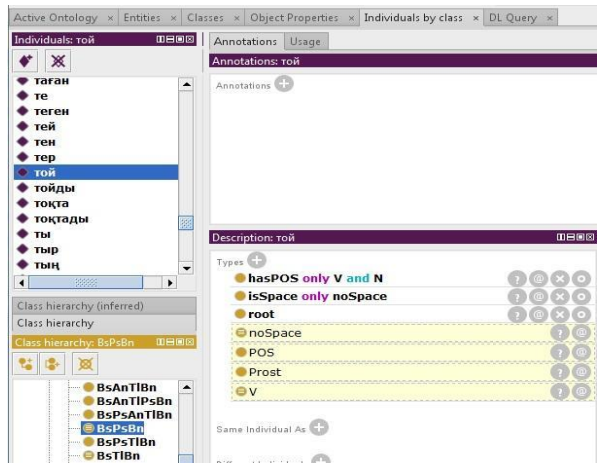

Figure 3. Definition of the word "toy" - "ate" as a non-spatial verb.

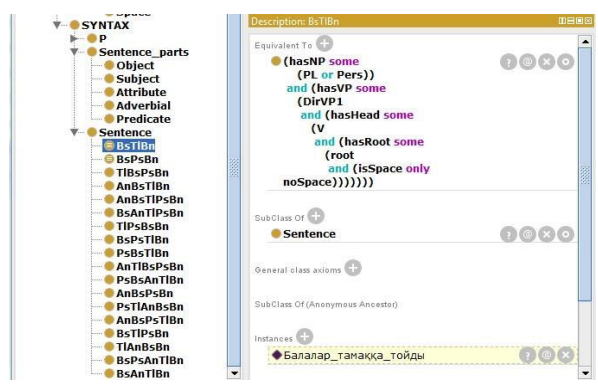

Figure 4. Ontological model of the Kazakh language sentence of the type BsTlBn.

Figure 5 shows the implementation of the Kazakh language sentence of the type BsPsBn (Bastauysh - Subject, Pysyktauysh - Adverbial, Bayandauysh - Predicate) in Protégé.

For this type of proposal, the following necessary and sufficient conditions must be met:

$$\mathrm{BsPsBn} \equiv \exists \mathrm{hasNP}(\mathrm{N\ or\ Pers}) \sqcap \exists \mathrm{hasVP}(\mathrm{DirVP1}(\sqcap\left(\exists \mathrm{hasHead}\left(\sqcap\left(\exists \mathrm{hasRoot}(\mathrm{root}\sqcap(\forall \mathrm{isSpace\ Space}))\right)\right)\right))$$

where: BsPsBn - Types of sentences consisting of Bastauysh - Subject, Pysyktauysh - Adverbial, Bayandauysh - Predicate;
hasNP - has a phrase;
N - a noun;
Pers - personal pronouns;
hasVP - has a verb phrase;
DirVP1 - verb management (noun in the directive case + verb)
hasHead - there is a main word;
V - the verb;
hasRoot - has a root;
root - root
isSpace - whether a word with a spatial meaning;
Space - spatial semantics.

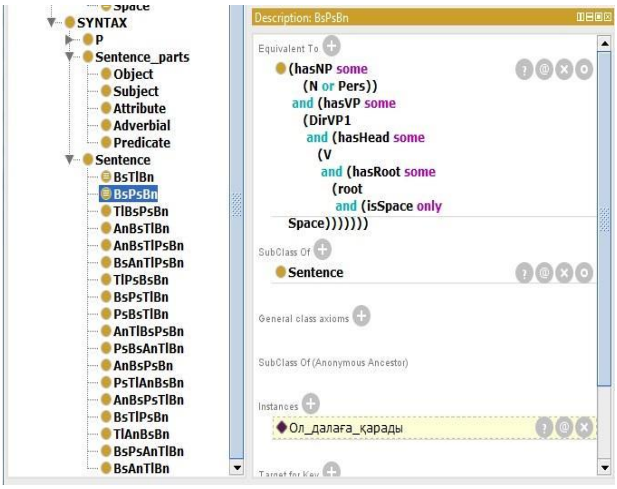

Figure 5. Ontological model of the Kazakh language sentence of the type BsPsBn.

Using these models, it is possible to automate the parsing of text and to determine the sentences parts of the Kazakh language. The parts of the sentences have a specific place in the sentence. You can not change the words in places as you want. When the place of the word changes their functions also change. For example, in this case, the subject can become another part of the sentence, and the functions of the other parts of the sentence will also change. At the same time, the main feature of the Kazakh language sentence structure is that the subject is at the beginning of the sentence, and the predicate is often used at the end. The Object stands before the predicate, and the Adverbial stands before the word relating to it. This structural model of the sentence is specific to the Kazakh language. However, the order of part of the sentences is also may change. For the Kazakh language the ways of making simple sentences had defined in [12, 13, 14]. Using these models, it is possible to automate the parsing of the text.

4. CONCLUSION

This article formalizes the syntactic rules of simple sentences of the Kazakh language using CFG and constructs ontological models of these rules taking into account the semantics of their components in the Protégé environment for syntax analysis. These formal models can be applied in computing applications, including grammar checking, semantic interpretation, dialogue understanding, and machine translation.

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